Mohammed Hossam Ahmed

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SUMMARY

Results-oriented Computer Science graduate and Software Developer with a strong foundation in backend development, data structures, and algorithms. Experienced in designing and delivering end-to-end software solutions, scalable RESTful APIs, microservices, and real-time systems using Node.js, TypeScript, Python (FastAPI), PostgreSQL, and MongoDB. Active open-source contributor to GitLab's core enterprise codebase (25+ Merge Requests) with hands-on experience in CI/CD, code reviews, and Agile workflows, competitive programmer (ICPC, Codeforces), and an experienced programming instructor who has taught 100+ students.

EDUCATION

Suez Canal University, Faculty of Computers and Information — **Ismailia, Egypt**
Bachelor's Degree in Computer Science (Class of 2026) — Aug 2022 – Jun 2026

OPEN SOURCE EXPERIENCE

Core Contributor | GitLab (gitlab-org/gitlab) — Open Source Development — **May 2026 – Present**

- Authored and submitted 25+ Merge Requests (MRs) to the core GitLab enterprise codebase, collaborating with maintainers through code reviews to enhance platform stability, component standardization, and API documentation.
- Resolved production, linting, and test spec issues across CI/CD, security, and analytics modules, including enforcing Vue.js/ESLint standards, fixing CI/CD pipeline failures, and correcting REST API curl commands in resource events.
- Awarded multiple Contributor Rewards from the GitLab Developer Relations team for consistent, impactful contributions and successful onboarding into the core ecosystem.

PROJECTS

Mindmate | Node.js, TypeScript, FastAPI, MongoDB, Socket.IO, Docker, Swagger, ESP32 — **Jan 2025 – Present**
github.com/MindMate-Project | alzaheimer-backend.onrender.com/api-docs

- Architected a multi-service backend linking a Node.js/TypeScript REST API with a Dockerized Python/FastAPI AI service (RetinaFace + ResNet50), supporting real-time bidirectional patient tracking via Socket.IO across custom ESP32 hardware.
- Designed a scalable NoSQL schema using Mongoose Discriminators to model a multi-role system (patient, caregiver, admin), with CRON-driven scheduling and Firebase push notification delivery to mobile clients.
- Enforced Role-Based Access Control (RBAC) across all caregiver-patient workflows via JWT middleware, with secure multimedia upload pipelines (Cloudinary, Multer) and interactive Swagger/OpenAPI documentation.

E-Commerce Backend System | Node.js, Express v5, MongoDB, JWT, Multer, bcrypt — **Sep 2024 – Dec 2024**
github.com/mohammedhossam86/Online-Shop | Live Demo

- Built a secure backend with a 3-tier RBAC system (Guest / Customer / Provider), enforcing route-level ownership checks and preventing cross-user data leakage at the database query layer.
- Implemented a full order lifecycle (pending to cancelled) with server-side validation of price, quantity, and stock, rejecting all client-sent pricing data, and scoped all cart and order queries strictly per authenticated user.
- Designed centralized error handling using custom AppError classes and async controller wrappers, eliminating unhandled promise rejections across the entire API surface.

SKILLS

Languages:	C++, JavaScript, TypeScript, Python, C, SQL
Backend:	Node.js, Express.js, FastAPI, REST API Design, Microservices, JWT, Socket.IO, Swagger (OpenAPI)
Databases:	PostgreSQL, MongoDB, Mongoose, SQLite
Tools & Platforms:	Git, GitHub, GitLab, Docker, Postman, CI/CD, Linux, Vercel, Render, Firebase, Cloudinary
Concepts:	Object-Oriented Programming (OOP), Data Structures, Algorithms, RBAC, MVC Architecture, SDLC, Agile
Soft Skills:	Problem-Solving, Technical Documentation, Code Reviews, Teamwork & Collaboration, Communication, Leadership

LEADERSHIP & TEACHING

Programming Instructor | Discovery Academy for Science and Technology **Sep 2026 – Present**

- Delivered 50+ sessions to 100+ students aged 5–18, tailoring curriculum to each age group: Scratch and LEGO-based computational thinking (ages 5–8), web development with HTML, CSS, and JavaScript (ages 8–12), and Python with problem solving (ages 12–18).
- Break down abstract concepts like loops, conditions, and algorithms into hands-on, age-appropriate activities that build logical thinking and confidence.

Mentor & IT Member at ICPC SCU, Suez Canal University **Jul 2024 – Present**

- Create programming problems and collaborate with the team to generate problem statements, test cases, and editorial solutions for university-level contests.
- Guide newcomers in competitive programming strategies and data structures through regular practice sessions, helping members qualify for regional rounds.

ACHIEVEMENTS

ICPC Regional Programming Contest

- Qualified to ECPC 2025, placing 17th on Day 8 of the ECPC Qualifications.
- Participated in ECPC Qualifications 2023 and 2024.
- Active competitive programmer on Codeforces with consistent contest participation.